

SAFETY DATA SHEET (GHS)

Cumene (Isopropylbenzene) | CAS: 98-82-8 | Formula: C9H12

PTT Phenol Train II (PPCL), Map Ta Phut, Thailand — HAZOP PSI Document | Issued: 2026-06-13 | SDS Version 1.0

DATA SOURCES CONSULTED

- ECHA Classification & Labelling Inventory — Cumene, CAS 98-82-8, harmonised classification (<https://echa.europa.eu/>)
- PubChem CID 7406 — GHS Classification (Rev.9), <https://pubchem.ncbi.nlm.nih.gov/compound/7406>
- ILO/NIOSH International Chemical Safety Card (ICSC) 0185 — Cumene
- NOAA CAMEO Chemicals — Cumene datasheet (<https://cameochemicals.noaa.gov/>)
- Accessed 2026-06-13

SECTION 1 — IDENTIFICATION

Product identifier: Cumene (Isopropylbenzene)

CAS Number: 98-82-8

Chemical formula: C9H12

Molecular weight: 120.19 g/mol

Synonyms: Isopropylbenzene; 2-Phenylpropane; (1-Methylethyl)benzene

Intended use: Primary process intermediate — product of the alkylation unit and feed to the oxidation (CHP formation) unit. The single highest-inventory organic stream on Train II. PTT Phenol Train II (PPCL), Map Ta Phut, Thailand

Emergency telephone: [Plant emergency response number — to be inserted by PPCL HSE]

SECTION 2 — HAZARD IDENTIFICATION

GHS classification: Flam. Liq. 3 (H226); Asp. Tox. 1 (H304); Skin Irrit. 2 (H315); Eye Irrit. 2 (H319); STOT SE 3 (H335); Aquatic Chronic 2 (H411)

Signal word: Danger

Hazard statements (H-codes): H226 (Flammable liquid and vapour); H304 (May be fatal if swallowed and enters airways); H315 (Causes skin irritation); H319 (Causes serious eye irritation); H335 (May cause respiratory irritation); H411 (Toxic to aquatic life with long lasting effects)

Precautionary statements (P-codes): P210, P233, P240, P241, P242, P243, P261, P264, P271, P273, P280, P301+P310, P303+P361+P353, P304+P340, P305+P351+P338, P331, P370+P378, P403+P233, P405, P501

GHS pictograms: GHS02 (Flammable), GHS07 (Irritant), GHS08 (Health hazard), GHS09 (Environmental hazard)

Other hazards: DATA QUALITY NOTE: a search result returned during compilation of this SDS suggested a 'Carcinogenicity Category 1B' classification for cumene; this is assessed as a probable data-aggregation artifact (cross-contamination with an unrelated substance on the same search results page) and has NOT been used in this SDS. The classification used here (Flam. Liq. 3, Asp. Tox. 1, Skin Irrit. 2, Eye Irrit. 2, STOT SE 3, Aquatic Chronic 2) reflects the ECHA harmonised classification for cumene. RECOMMEND a manual cross-check of the current ECHA C&L Inventory entry for CAS 98-82-8 before finalising the HAZOP PSI dossier. As the single largest organic inventory item in the process (alkylation product / oxidation feed), even a moderate-severity classification carries significant aggregate risk due to volume.

SECTION 3 — COMPOSITION / INFORMATION ON INGREDIENTS

Cumene: CAS 98-82-8 — process-grade purity typically >99.5%; confirm against PPCL alkylation product / oxidation feed specification (trace benzene, di-isopropylbenzenes, and n-propylbenzene possible)

SECTION 4 — FIRST AID MEASURES

Inhalation: Remove to fresh air. If breathing is difficult, administer oxygen; perform artificial respiration if breathing has stopped. Obtain medical attention — vapour can cause respiratory tract irritation (H335) and CNS depression (headache, dizziness) at higher concentrations.

Skin contact: Remove contaminated clothing. Wash skin thoroughly with soap and water. Obtain medical attention if irritation persists or develops.

Eye contact: Flush immediately with copious water for at least 15 minutes, holding eyelids open. Obtain medical attention if irritation persists.

Ingestion: Do NOT induce vomiting — aspiration hazard (H304); aspiration into the lungs can cause chemical pneumonitis and may be fatal. Obtain immediate medical attention. If vomiting occurs spontaneously, keep head below hips.

Notes to physician: Monitor for chemical pneumonitis following ingestion/aspiration; treat respiratory irritation symptomatically; CNS effects from inhalation are generally reversible on removal from exposure

SECTION 5 — FIREFIGHTING MEASURES

Suitable extinguishing media: Dry chemical, CO₂, alcohol-resistant foam, water spray/fog for cooling

Unsuitable extinguishing media: High-volume straight water streams directly onto burning liquid (may spread fire — cumene is less dense than water)

Special hazards: Flammable liquid and vapour (H226) — flash point ~31-44°C depending on test method/source (see Section 9; confirm against this facility's process operating temperatures, which may approach or exceed the flash point in some areas). Vapours heavier than air can accumulate and travel to ignition sources. As the highest-inventory organic stream, a release represents a major pool-fire/vapour-cloud fuel source.

Special protective equipment for firefighters: Full bunker gear with positive-pressure SCBA; cool fire-exposed vessels/pipework with water spray from a safe distance; approach from upwind

Hazardous combustion products: Carbon monoxide, carbon dioxide, dense smoke

SECTION 6 — ACCIDENTAL RELEASE MEASURES

Personal precautions: Eliminate ignition sources. Evacuate non-essential personnel. Wear appropriate respiratory protection for vapour exposure given STOT SE 3 (respiratory irritation) classification. Bond/ground equipment during transfer.

Environmental precautions: Prevent entry to drains, sewers, watercourses and soil — toxic to aquatic life with long lasting effects (H411)

Containment and cleanup: Contain spill with non-combustible absorbent (sand, vermiculite, commercial sorbent). Collect in closed, labelled containers. Ventilate area; given the large potential inventory of this stream, large releases should trigger the site's major-spill/fire response procedures.

Reference to other sections: See Section 7 (Handling/Storage), Section 8 (PPE), Section 9 (flammability data)

SECTION 7 — HANDLING AND STORAGE

Handling precautions: Handle in closed systems; use explosion-proof/intrinsically safe electrical equipment in classified areas. Bond and ground transfer lines/containers. Avoid generating vapours/mists; ensure adequate ventilation at sample points.

Storage conditions: Store in closed tanks in a well-ventilated area, away from heat, sparks, open flames and other ignition sources. As the largest organic inventory in Train II, storage tank fire protection (foam systems, bunding/secondary containment, fixed water spray cooling) should be confirmed against the facility's fire risk assessment.

Temperature limits: Maintain storage temperatures below the flash point with adequate margin; confirm operating temperatures of cumene-containing vessels do not approach the flash point range during normal/upset conditions

Pressure: Atmospheric/low-pressure storage with conservation venting to vapour recovery/flare as required

Segregation requirements: Segregate from strong oxidisers and sources of ignition. Note: cumene is the feedstock for the oxidation reactors where it is deliberately reacted with oxygen/air to form CHP — segregation requirements in storage/transfer areas differ from those in the oxidation reaction section itself (see CHP SDS, CAS 80-15-9, for oxidation-section-specific guidance).

Special requirements: LEL monitoring in storage and transfer areas; banded secondary containment for storage tanks given the large inventory

Specific end-use recommendations: Confirm trace impurity profile (especially di-isopropylbenzenes and residual benzene) against the oxidation reactor feed specification, as impurities can affect CHP formation selectivity and downstream decomposition chemistry

SECTION 8 — EXPOSURE CONTROLS / PERSONAL PROTECTION

Occupational Exposure Limits (OEL): ACGIH TLV-TWA: 50 ppm (with Skin notation); OSHA PEL: 50 ppm TWA. Thailand OEL: confirm current Ministry of Labour value (commonly aligned to 50 ppm TWA).

TWA: 50 ppm (ACGIH/OSHA)

STEL: Not formally established by ACGIH

Ceiling: Not established

Skin notation: Yes — dermal absorption contributes to systemic exposure

IARC/NTP/OSHA carcinogen status: Not currently classified as a carcinogen by IARC (Group 3 — not classifiable) per most recent assessments; see Section 2 data-quality note regarding a Carc. 1B search result that was NOT used in this SDS pending manual verification

Engineering controls: Closed system handling; local exhaust ventilation at sample points and open transfer operations; LEL monitoring

PPE — Respiratory: Air-purifying respirator with organic vapour cartridge for tasks with vapour exposure potential; SCBA for tank entry or major release response

PPE — Gloves: Chemical-resistant gloves (e.g. nitrile, Viton) — verify breakthrough time for cumene

PPE — Eyes: Chemical splash goggles; face shield for splash-potential tasks

PPE — Body: Flame-resistant (FR), chemical-resistant coveralls; antistatic footwear in classified areas

Biological monitoring: Not routinely established

SECTION 9 — PHYSICAL AND CHEMICAL PROPERTIES

Appearance: Colourless liquid

Odour: Sharp, penetrating, gasoline-like aromatic odour

Odour threshold: ~0.01-0.1 ppm (low — detectable well below irritant concentrations)

pH: Not applicable (organic liquid, immiscible with water)

Melting point / Freezing point: -96°C

Boiling point (at 1 atm): 152.4°C

Flash point: 31-44°C depending on test method/source (closed cup values around 31-39°C are most commonly cited; some sources report up to 44°C) — EXPERIMENTAL VALUE TO BE CONFIRMED against a single authoritative source (e.g. ECHA dossier) for the HAZOP PSI dataset, as this directly affects classified-area and process-temperature risk screening for the largest-inventory stream

Evaporation rate: Slower than benzene/cumene-lighter aromatics (lower vapour pressure); quantitative rate not commonly reported

Flammability (solid/gas): Flammable liquid

Flammability limits in air: LEL 0.8 vol%; UEL 6.0 vol%

Vapour pressure (at 20°C): ~0.45-0.6 kPa (~3.4-4.5 mmHg)

Vapour density (air = 1): ~4.1 (heavier than air)

Relative density (water = 1): ~0.862 (at 20°C)

Solubility (water): ~50-60 mg/L at 25°C (practically insoluble)

Partition coefficient (log Kow): ~3.66

Auto-ignition temperature: ~424°C

Decomposition temperature: Not applicable under normal use; thermal decomposition at very high temperatures yields typical aromatic combustion products

Viscosity: ~0.75 mPa·s at 25°C

Explosive properties: Not classified as explosive substance; forms flammable vapour/air mixtures within LEL-UEL range

Oxidising properties: Not an oxidiser. NOTE: cumene is deliberately oxidised (autooxidation with air/O₂) in the process oxidation reactors to form cumene hydroperoxide (CHP, CAS 80-15-9) — see CHP SDS for the hazards of this reaction product

Molecular weight: 120.19 g/mol

SECTION 10 — STABILITY AND REACTIVITY

Reactivity: Stable under normal ambient conditions; readily undergoes autooxidation in the presence of oxygen/air, heat and/or light to form cumene hydroperoxide (CHP) — this is the basis of the intended process oxidation step but represents

an unwanted reaction if it occurs uncontrolled in storage/transfer (e.g. prolonged air exposure of warm cumene in tanks)

Chemical stability: Stable under recommended storage conditions; commercial cumene may contain a peroxidation inhibitor — confirm inhibitor presence/concentration for storage tanks not under inert gas blanketing

Possibility of hazardous reactions: Autoxidation to CHP (and other peroxides) on prolonged exposure to air, especially at elevated temperature or under light — accumulated peroxides in stored cumene are a recognised hazard (shock/heat-sensitive peroxide residues on evaporation); reacts with strong oxidisers; can react with strong acids under certain conditions

Conditions to avoid: Prolonged storage in contact with air without inert gas blanketing or inhibitor, especially at elevated temperature or in sunlight — promotes peroxide (CHP) formation; heat, sparks, open flames

Incompatible materials: Strong oxidisers; conditions promoting autoxidation (uninhibited storage with air ingress)

Hazardous decomposition products: Combustion produces CO, CO₂; autoxidation produces cumene hydroperoxide (CHP) and related peroxide species — see CHP SDS (CAS 80-15-9) for decomposition hazards of this product

SECTION 11 — TOXICOLOGICAL INFORMATION

Routes of exposure: Inhalation, dermal contact, ingestion (aspiration hazard)

Acute toxicity — Oral LD₅₀: ~1,400-2,900 mg/kg, rat (literature range)

Acute toxicity — Dermal LD₅₀: >3,000-12,800 mg/kg, rabbit (low acute dermal lethality)

Acute toxicity — Inhalation LC₅₀: ~8,000 ppm (4 h), rat (approximate literature value)

Skin corrosion/irritation: Causes skin irritation (H315); defatting on repeated contact

Serious eye damage/irritation: Causes serious eye irritation (H319)

Respiratory/skin sensitisation: Not classified as a sensitiser

Germ cell mutagenicity: Not classified

Carcinogenicity: Not classified by IARC (Group 3 — not classifiable as to carcinogenicity to humans) per most recent assessments. See Section 2 data-quality note: a 'Carc. 1B' result encountered during research was NOT applied pending manual verification of the source.

Reproductive toxicity: Not classified

STOT (single exposure): May cause respiratory irritation (H335); high vapour concentrations can cause CNS depression (headache, dizziness, drowsiness)

STOT (repeated exposure): Not classified

Aspiration hazard: May be fatal if swallowed and enters airways (H304) — chemical pneumonitis risk

SECTION 12 — ECOLOGICAL INFORMATION

Aquatic toxicity (LC₅₀/EC₅₀): Fish LC₅₀ (96h) ~2-10 mg/L (species dependent); Daphnia EC₅₀ (48h) ~2-4 mg/L — Toxic to aquatic life with long lasting effects (H411)

Persistence and degradability: Expected to biodegrade under aerobic conditions; not considered highly persistent, but moderate log K_{ow} gives some potential for sorption to sediment/soil

Bioaccumulative potential: Moderate (log K_{ow} ~3.66) — some potential for bioaccumulation, though generally not considered significant

Mobility in soil: Moderate mobility; low water solubility favours sorption to organic matter in soil

Other adverse effects: No known ozone-depletion potential; VOC — relevant to air emissions permitting

SECTION 13 — DISPOSAL CONSIDERATIONS

Disposal methods: Dispose of via licensed hazardous waste incineration or other approved method in accordance with Thailand Hazardous Substance Act B.E. 2535 and site waste management procedures; do not discharge to drains/sewers

Contaminated packaging: Drain thoroughly; dispose of as hazardous waste or return for reconditioning per supplier program. CAUTION: containers/equipment that have held cumene for extended periods may contain accumulated peroxide (CHP) residues — treat per CHP SDS precautions before cutting/welding or other hot work.

Regulatory reference: Thailand Hazardous Substance Act B.E. 2535

SECTION 14 — TRANSPORT INFORMATION

UN Number: UN 1918

Proper shipping name: ISOPROPYLBENZENE (CUMENE)

Transport hazard class(es): Class 3 (Flammable liquid)

Packing group: III

Environmental hazards (marine pollutant): Yes — marine pollutant (Aquatic Chronic 2)

Special precautions for user: Ensure bonding/grounding during transfer; verify absence of significant peroxide accumulation before transferring aged material

Bulk transport (MARPOL/IBC): Classify per IBC Code if shipped in bulk by sea

SECTION 15 — REGULATORY INFORMATION

Safety and health regulations: See country-specific notes below

Thailand: Hazardous Substance Act B.E. 2535 — classification depends on concentration/use; verify listing (List 1-4) with Department of Industrial Works.

EU: Registered under REACH (Regulation (EC) No 1907/2006); classified under CLP Regulation (EC) No 1272/2008.

US: Subject to OSHA Hazard Communication Standard 29 CFR 1910.1200 (HazCom 2012, aligned with GHS Rev.7).

ASEAN: Classification consistent with ASEAN GHS implementation (harmonised with UN GHS).

Chemical safety assessment: Yes — performed by upstream registrants under REACH; site-specific CSA recommended as part of PPCL PSI documentation.

SECTION 16 — OTHER INFORMATION

Revision date: 2026-06-13

Revision number: 1.0

Key changes from previous version: Initial issue — compiled for HAZOP PSI (CAS 98-82-8). NOTE: a 'Carcinogenicity Category 1B' result returned by one search source was assessed as a likely data-aggregation artifact and was NOT incorporated — RECOMMEND manual verification against the current ECHA C&L Inventory before finalising the dossier.

Data sources used in this SDS: See 'Data Sources Consulted' block at top of document

Disclaimer: This SDS was compiled for HAZOP Process Safety Information purposes at PTT Phenol Train II (PPCL), Map Ta Phut, Thailand. Data sourced from published authoritative references (ECHA, PubChem/GHS Rev.9, NIOSH, NTP, supplier SDSs, CAMEO Chemicals, peer-reviewed literature). Verify against current manufacturer SDS before use for regulatory compliance or labelling purposes.

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